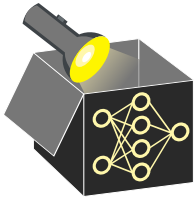
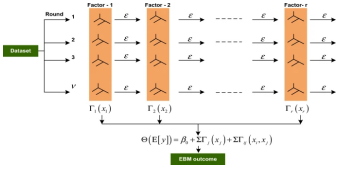
# Interpretable Machine Learning



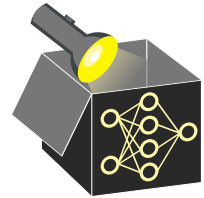
# Explainable Boosting Machines




## Learning goals

- Motivation from GAM
- Intelligible GAM
- Accurate GAM + Pairwise Interactions
- FAST feature interaction detection

# GENERALIZED ADDITIVE MODELS

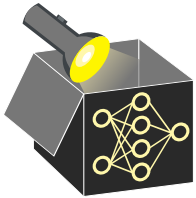


**Recall idea of GAMs**:

$$g(\mathbb{E}(y_i \mid \mathbf{x}_i)) = \theta_0 + f_1(x_{i1}) + f_2(x_{i2}) + \ldots + f_p(x_{ip}), \quad i = 1, \cdots, N$$

- One shape function (spline) $f_j$ for each feature $\rightsquigarrow$ Intelligible
- Including non-linear effects
- Better than GLMs in accuracy
- **Explainable Boosting Machines** $\leftarrow$ GAM + Gradient Boosting + Tree ensembles ▸ Lou et al. 2012
  $\rightsquigarrow$ More intelligible than unrestricted full complexity models (e.g. random forest)
  + High accuracy close to unrestricted full complexity models

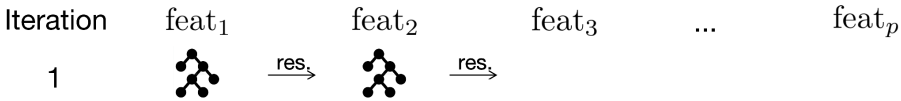
# INTELLIGIBLE GAM - ALGORITHM SKETCH - PART I

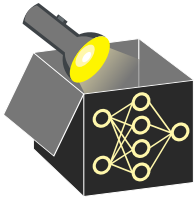


| Iteration | $\text{feat}_1$ | | $\text{feat}_2$ | $\text{feat}_3$ | ... | $\text{feat}_p$ |
|---|---|---|---|---|---|---|
| 1 | | $\xrightarrow{\text{res}_1}$ | | | | |

- Set all shape functions $f_j^{(0)}$ to zero, $j = 1, \cdots, p$
- Calculate the initial pseudo-residual (i.e. the negative gradient of the loss function w.r.t. the current model prediction), e.g.
  1. for squared error loss: $\tilde{r}_i^{(0)} = -\frac{\partial L}{\partial \hat{y}_i} = y_i - \hat{y}_i^{(0)} = y_i - \sum_j f_j^{(0)}(x_{ij})$
  2. for log loss: $\tilde{r}_i^{(0)} = -\frac{\partial L}{\partial \hat{y}_i} = y_i - \sigma(\hat{y}_i^{(0)})$ with $\sigma(\cdot)$ being the sigmoid function
- Fit a shallow bagging tree with 2-4 leaves $S_1^{(1)} : x_{i1} \to \tilde{r}_i$ **only using the first feature** to predict the pseudo-residuals, with $\{x_{i1}, \tilde{r}_i\}_1^N$ as training dataset
- Update $f_1$: $f_1^{(1)}(x_{i1}) = f_1^{(0)} + \eta \cdot S_1^{(1)}(x_{i1})$
- Update $\hat{y}_i$: $\hat{y}_i^{(1)} = \sum_j f_j^{(1)}(x_{ij})$
- Update the residual w.r.t. the original target, e.g. $\tilde{r}_i^{(1)} = y_i - \hat{y}_i^{(1)}$lime
- Go to the second feature

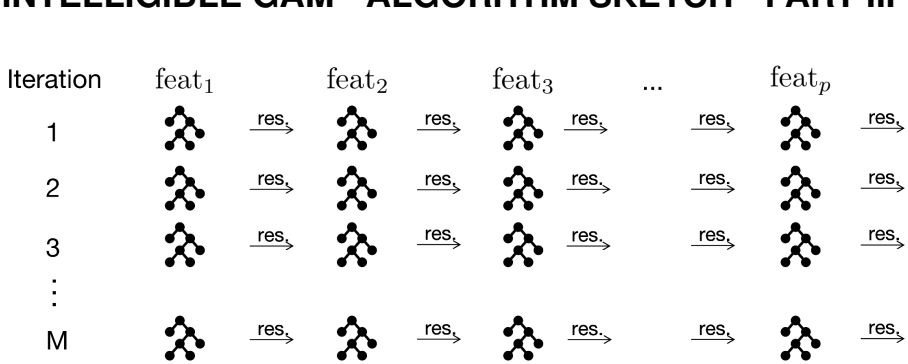
# INTELLIGIBLE GAM - ALGORITHM SKETCH - PART II

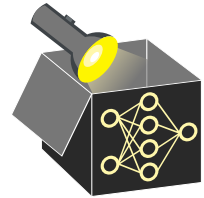


- Fit a shallow bagging tree $S_2^{(1)}$ **only using the second feature** to predict the updated residuals
- Update $f_2$ with $f_2^{(1)}(x_{i2}) = f_2^{(0)} + \eta \cdot S_2^{(1)}(x_{i2})$ and then update $\hat{y}_i^{(1)}$
- Update the residual: this residual takes into account both what was learned on the first tree and second tree for the first two features
- Go to $\text{feat}_3$

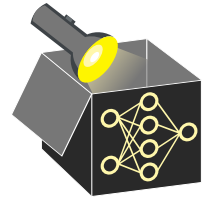
# INTELLIGIBLE GAM - ALGORITHM SKETCH - PART III



- Go through every feature and keep updating the residual
- Grow a small tree on each feature but just one at a time in each iteration
- Do $M$ iterations through these features until convergence (e.g. $M = 10000$)
  $\rightsquigarrow$ keep the learning rate $\eta$ so small that feature ordering does not matter

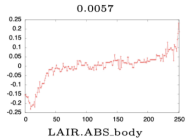
# INTELLIGIBLE GAM - PREDICTION



- Output: $M$ trees which were only trained on $\text{feat}_1$
  + $M$ trees which were only trained on $\text{feat}_2$
  $\dots$
  + $M$ trees which were only trained on $\text{feat}_p$
- Summarize the weighted predictions of $M$ trees for each unique value of $\text{feat}_1$
  $\hat{f}_1(x_1) = \sum_{m=1}^{M} \eta \cdot S_1^{(m)}(x_1)$, and plot $\hat{f}_1(x_1)$ vs. $x_1$ as a 2D graph
- Generate a 2D graph for every feature in the same way

$\rightsquigarrow$ Intelligible GAM: A series of 2D graphs as a perfect summary of predictions

Prediction $=$

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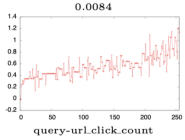


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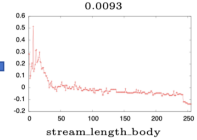
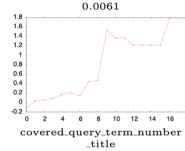
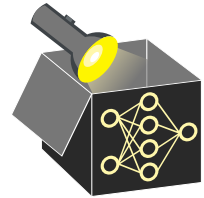



$+$ $\dots$ $+$

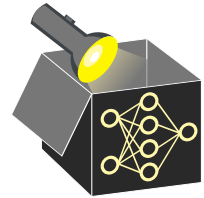


# INTELLIGIBLE GAM - NOVELTY AND EXTENSION



- In each boosting step, use a bagging tree limited in size (2-4 leaves)
  - High accuracy, low variance, prevent overfitting
  - Provides discrete shape functions
  - Captures missing information in spline plots
  - High intelligibility
- Not as accurate as unrestricted full complexity model
- In the next step: ▸ Lou et al. 2013
  - Add pairwise feature interaction terms to the model
  - Replace the 2-4 regions of small tree
  - By a dynamic programming algorithm called FAST

# ACCURATE GAM PLUS PAIRWISE INTERACTIONS

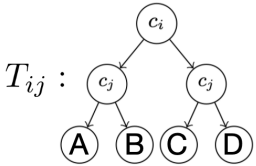


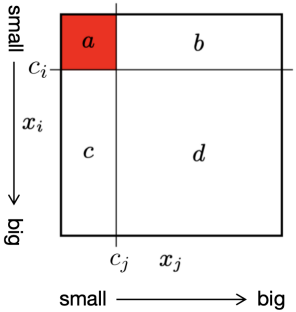
**Generalized Additive Models plus Interactions**:

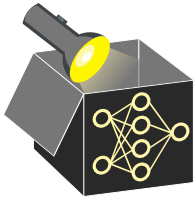
$$g(\mathbb{E}(y \mid \mathbf{x})) = \theta_0 + \sum f_i(x_i) + \sum f_{ij}(x_i, x_j) \quad \textit{for} \quad i, j = 1, \cdots, p, \quad i \neq j$$

- $O(p^2)$: a large number of pairwise interactions if $p$ is big
  $\rightsquigarrow$ Select only a small number of the most important interactions
- Intuitively, significant reduction in RSS $\rightarrow$ Strong pair interaction
- **FAST** algorithm does quick ranking without fully building every interaction function

# FAST FEATURE INTERACTION DETECTION - PART I



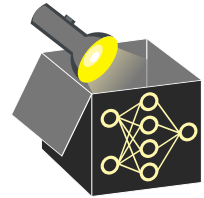



- $T_{ij}$: A tree-like interaction predictor with 4 leaves for each pair of features
- $c_i$, $c_j$: One cut (grid point) on the **sorted** values of feature $x_i$ and $x_j$, respectively
- $A$, $B$, $C$, $D$: Notation for 4 leaf regions
- **Idea**: Search for all possible $(c_i, c_j)$ and pick the best $T_{ij}$ with the lowest RSS
- Prediction value on region r, $r \in \{A, B, C, D\}$:

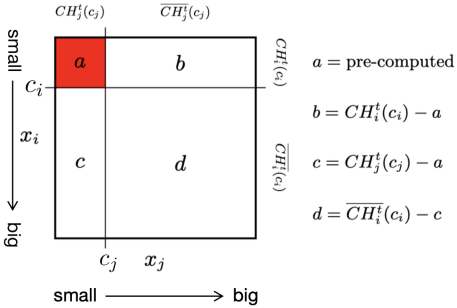
$$\hat{y}_r = T_{ij}.r = \frac{1}{|r|} \sum_{(\mathbf{x},y)\in r} y = \frac{\text{Sum of targets in } r}{\text{Sum of counts in } r}$$

- $$\begin{aligned}\text{RSS} &= \textstyle\sum_{k=1}^{N} (y_k - T_{ij}(\mathbf{x}_k))^2 \\ &= \textstyle\sum_{k=1}^{N} y_k^2 - 2\sum_r \left(\hat{y}_r \cdot \sum_{(\mathbf{x},y)\in r} y\right) \\ &\quad + \textstyle\sum_r \hat{y}_r^2 \cdot |r|\end{aligned}$$

# FAST FEATURE INTERACTION DETECTION - PART II



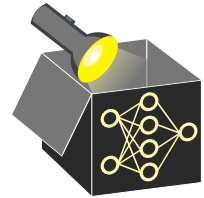
- How to quickly get sum of targets and sum of counts for all possible cuts?
  $\rightsquigarrow$ Use **marginal cumulative histograms** (CH)




An illustration for computing sum of targets for each quadrant

- For example:
  $CH_j^t(c_j)$: cumulative sum of targets for data points with $x_j \leq c_j$;
  $\overline{CH_j^t}(c_j)$: cumulative sum of targets for data points with $x_j > c_j$
- Create **two lookup tables** to store values for every $(c_i, c_j)$, :
  1. Compute the cumulative sum of targets in four regions, and store the values in the **table of targets** $L^t(c_i, c_j) = [a, b, c, d]$
  2. Compute the cumulative sum of counts in four regions, and store the values in the **table of counts** $L^w(c_i, c_j) = [a, b, c, d]$
- Given any $(c_i, c_j)$, values of $(b, c, d)$ can be easily derived with known $a$
  $\rightsquigarrow$Construct lookup tables using dynamic programming

# FAST FEATURE INTERACTION DETECTION - PART III



- For $x_i$ with $d_i$ possible values and $x_j$ with $d_j$ possible values:
  $\rightsquigarrow$ compute and store $d_i d_j$ records of $[a, b, c, d]$ in both table of targets and table of counts
- For each pair of $(x_i, x_j)$, calculate RSS for each possible $(c_i, c_j)$ and pick the best $T_{ij}$ with the lowest RSS [Note: $\hat{y}_A$ can be simply calculated as $L^t(c_i, c_j).a / L^w(c_i, c_j).a$]

$d_i d_j$ possible $(c_i, c_j)$

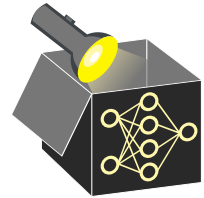
| | | | | | | |
|---|---|---|---|---|---|---|
| a | b | $\Rightarrow$ | $\hat{y}_A$ | $\hat{y}_B$ | $\Rightarrow$ | $RSS_1$ |
| c | d | | $\hat{y}_C$ | $\hat{y}_D$ | | |
| a | b | $\Rightarrow$ | $\hat{y}_A$ | $\hat{y}_B$ | $\Rightarrow$ | $RSS_2$ |
| c | d | | $\hat{y}_C$ | $\hat{y}_D$ | | |
| $\vdots$ | | | $\vdots$ | | | $\vdots$ |
| a | b | $\Rightarrow$ | $\hat{y}_A$ | $\hat{y}_B$ | $\Rightarrow$ | $RSS_{d_i d_j}$ |
| c | d | | $\hat{y}_C$ | $\hat{y}_D$ | | |

select the lowest RSS and assign it as the weight for $(x_i, x_j)$ to measure the strength of interaction

# FAST FEATURE INTERACTION DETECTION - PART IV

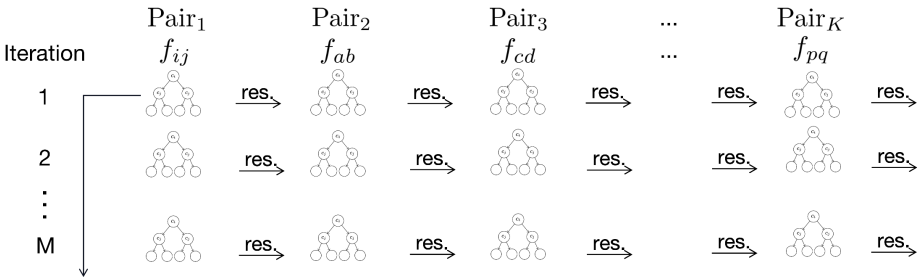
- Complexity analysis:
  Computing cumulative histograms: $O(N)$ ($N$: number of observations)
  Constructing matrices of sum values: $O(N + d_i d_j)$
  $\rightsquigarrow$ Complexity per pair: $O(N + d_i d_j)$
- Discretize continuous feature values in $b$ bins ($b \leq 256$)
  $\rightsquigarrow$ $O(N + b^2)$ per pair
  $\rightsquigarrow$ Choose optimal number of bins $b$ w.r.t. bias-variance trade-off
- Quick ranking and select top K pairs

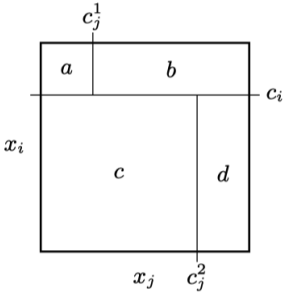
# ACCURATE GAM - TWO-STAGE CONSTRUCTION

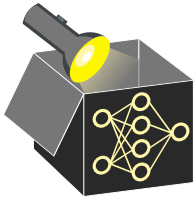


1. Fit main effects
   - Fit model only using univariate terms
   - Fix the main effects model
2. Fit pairwise residual of main effects model to original targets
   - Use FAST to sort through $O(p^2)$ pairs
   - User selects number $K$ of pairs to be added
   - Run boosting algorithm to fit $K$ pairs
   - Final model = $p$ mains + $K$ pairs

# ACCURATE GAM - 2. STAGE ALGORITHM SKETCH

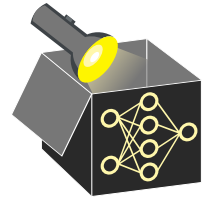


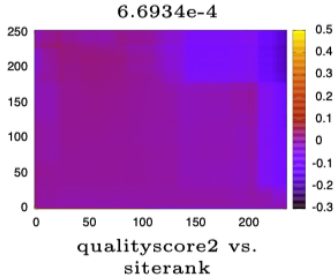



- To model the pairwise interactions effectively and efficiently ⇝Replace the shallow tree ensembles used in Intelligible GAM with a tree-like predictor (similar to FAST)
- 3 cuts instead of 2 cuts
- Re-use the stored values in the lookup tables for sum of targets and sum of counts from FAST
- Greedy search for the best combination of cuts (lowest RSS)

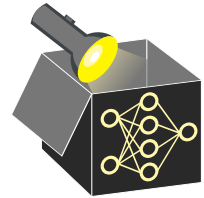
# ACCURATE GAM - PREDICTION



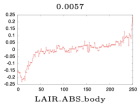
- Output: $M$ predictors which were only trained on $\text{Pair}_1$
  + $M$ predictors which were only trained on $\text{Pair}_2$
  . . .
  + $M$ predictors which were only trained on $\text{Pair}_K$
- Summarize the weighted predictions of $M$ models in a 3D heatmap
- The dimension of color represents the contribution of pairwise feature values to the prediction
- Generate a 3D heatmap for each pair of features



# EBM - FINAL MODEL



$\text{Main}_1$ $\text{feat}_1$ | $\text{Main}_2$ $\text{feat}_2$ | $\text{Main}_3$ $\text{feat}_3$ | ... ... | $\text{Main}_p$ $\text{feat}_p$

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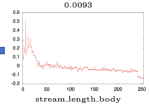


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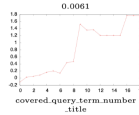


Prediction =

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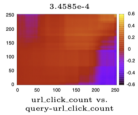


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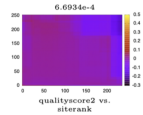


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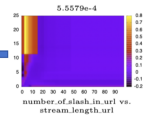


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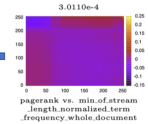


$\text{Pair}_1$ $f_{ij}$ | $\text{Pair}_2$ $f_{ab}$ | $\text{Pair}_3$ $f_{cd}$ | ... ... | $\text{Pair}_K$ $f_{pq}$